Kubernetes and Cloud Native Associate - KCNA

Scheduling

Configuring Kubernetes Scheduler Profiles

In this article, we explore scheduler profiles and the inner workings of the Kubernetes scheduler using a practical example where a Pod is scheduled to one of four nodes in a Kubernetes cluster.

Pod Definition Example

Below is an example of a Pod definition file. This Pod requires 10 CPU units and will only be scheduled on a node that meets or exceeds that capacity.

```yaml
apiVersion: v1
kind: Pod
metadata:
  name: simple-webapp-color
spec:
  priorityClassName: high-priority
  containers:
    - name: simple-webapp-color
      image: simple-webapp-color
      resources:
        requests:
          memory: "1Gi"
          cpu: 10
```

Each node in the cluster has a defined amount of available CPU. As Pods are created, they enter a scheduling queue where they are arranged based on the priority specified in their configuration. In this scenario, our Pod is assigned a high priority by using a PriorityClass object. Here is an example of how to create such a PriorityClass:

```
apiVersion: scheduling.k8s.io/v1
kind: PriorityClass
metadata:
  name: high-priority
value: 1000000
globalDefault: false
description: "This priority class should be used for XYZ service pods only."
```

Note

High-priority settings ensure that Pods with this classification are placed at the front of the scheduling queue.

Scheduling Phases

The Pod scheduling process comprises three main phases:

1. **Filter Phase:**
   In this phase, the scheduler eliminates nodes that do not satisfy the Pod's resource requirements. For instance, if the first two nodes do not have the needed 10 CPU units available, they are filtered out.

2. **Scoring Phase:**
   Nodes that pass the filter phase are then scored. The scheduler assigns each node a score based on factors such as the remaining CPU after allocating the Pod's requirements. For example, if one node has 2 CPU units remaining while another has 6, the latter will receive a higher score.

3. **Binding Phase:**
   In the final phase, the Pod is assigned to the node with the best score during the binding process.

Key Plugins in the Scheduling Process

Plugins are integral to the Kubernetes scheduling process. Here are some examples:

- **Priority Sort Plugin:**
  During the scheduling queue phase, this plugin orders Pods based on their assigned priority.

- **Node Resources Fit Plugin:**
  This plugin is active during the filter phase to exclude nodes lacking sufficient resources. Additionally, during the scoring phase, this plugin re-evaluates nodes based on free resources.

- **Node Unschedulable Plugin:**
  This plugin ensures that nodes marked as unschedulable do not have Pods assigned. For example, running the command:

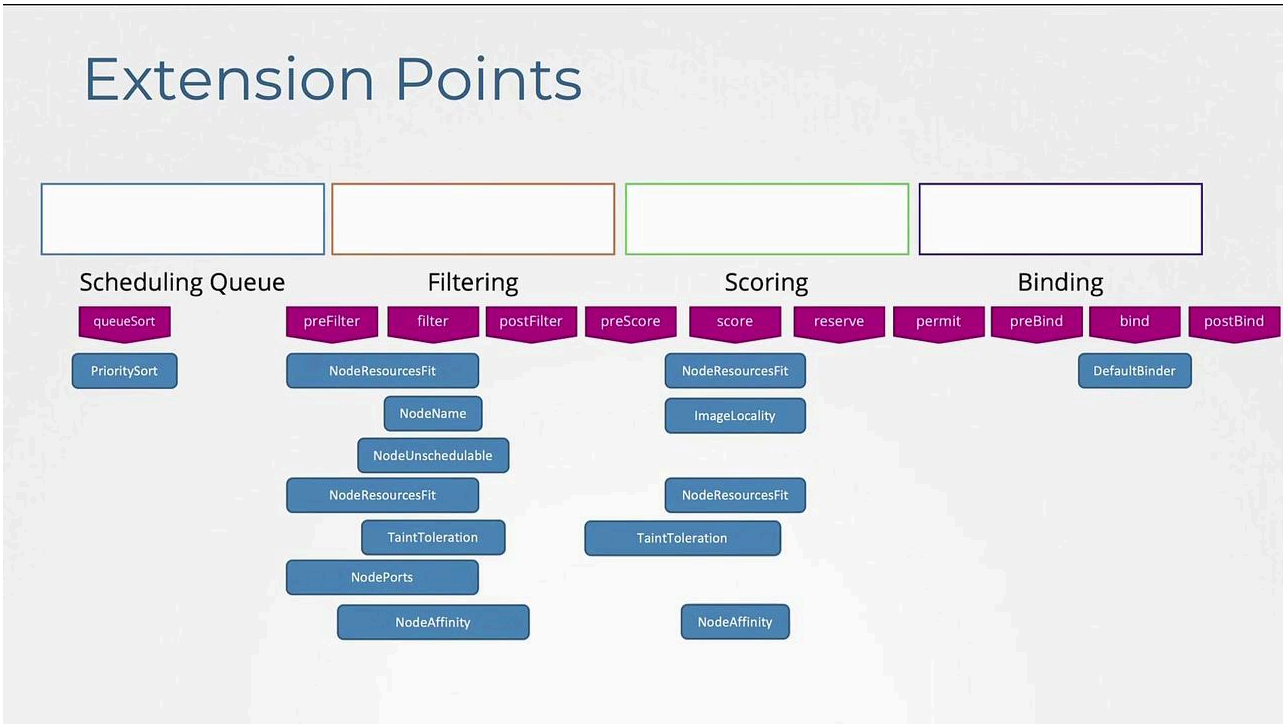
```
controlplane ~ ➜ kubectl describe node controlplane
Name:               controlplane
Roles:              control-plane
CreationTimestamp:  Thu, 06 Oct 2022 06:19:57 -0400
Taints:             node.kubernetes.io/unschedulable:NoSchedule
Unschedulable:      true
Lease:
```

  confirms that the node unschedulable plugin prevents Pod scheduling on such nodes.

- **Image Locality Plugin:**
  This plugin is a soft preference during the scoring phase, favoring nodes that already contain the required container image.

- **Default Binder Plugin:**
  In the binding phase, this plugin finalizes the Pod-to-node assignment.

Kubernetes' extensible design lets you customize active plugins at each extension point, including pre-filter, filter, post-filter, pre-score, score, reserve, pre-bind, and post-bind. You can also integrate custom plugins to meet specific requirements.



Using Multiple Scheduling Profiles

Kubernetes' extensibility is further demonstrated by its support for multiple scheduler profiles within a single scheduler binary. This feature, introduced in Kubernetes 1.18, simplifies process maintenance and reduces race conditions by eliminating the need for separate scheduler binaries (such as default scheduler, my-scheduler, and my-scheduler2).

Consider the following configuration files that define separate scheduler configurations with unique scheduler names:

```
# my-scheduler-2-config.yaml
apiVersion: kubescheduler.config.k8s.io/v1
kind: KubeSchedulerConfiguration
```

```
profiles:
- schedulerName: my-scheduler-2
```

```
# my-scheduler-config.yaml
apiVersion: kubescheduler.config.k8s.io/v1
kind: KubeSchedulerConfiguration
profiles:
- schedulerName: my-scheduler
```

```
# scheduler-config.yaml
apiVersion: kubescheduler.config.k8s.io/v1
kind: KubeSchedulerConfiguration
profiles:
- schedulerName: default-scheduler
```

Each scheduler profile functions as an independent scheduler within the same binary. To further customize these profiles, you can manipulate the plugin settings by disabling default plugins or enabling custom ones. Below is a sample configuration showcasing these customizations:

```
apiVersion: kubescheduler.config.k8s.io/v1
kind: KubeSchedulerConfiguration
profiles:
- schedulerName: my-scheduler-2
  plugins:
    score:
      disabled:
      - name: TaintToleration
      enabled:
      - name: MyCustomPluginA
      - name: MyCustomPluginB

- schedulerName: my-scheduler-3
  plugins:
    preScore:
      disabled:
      - name: "*"
```

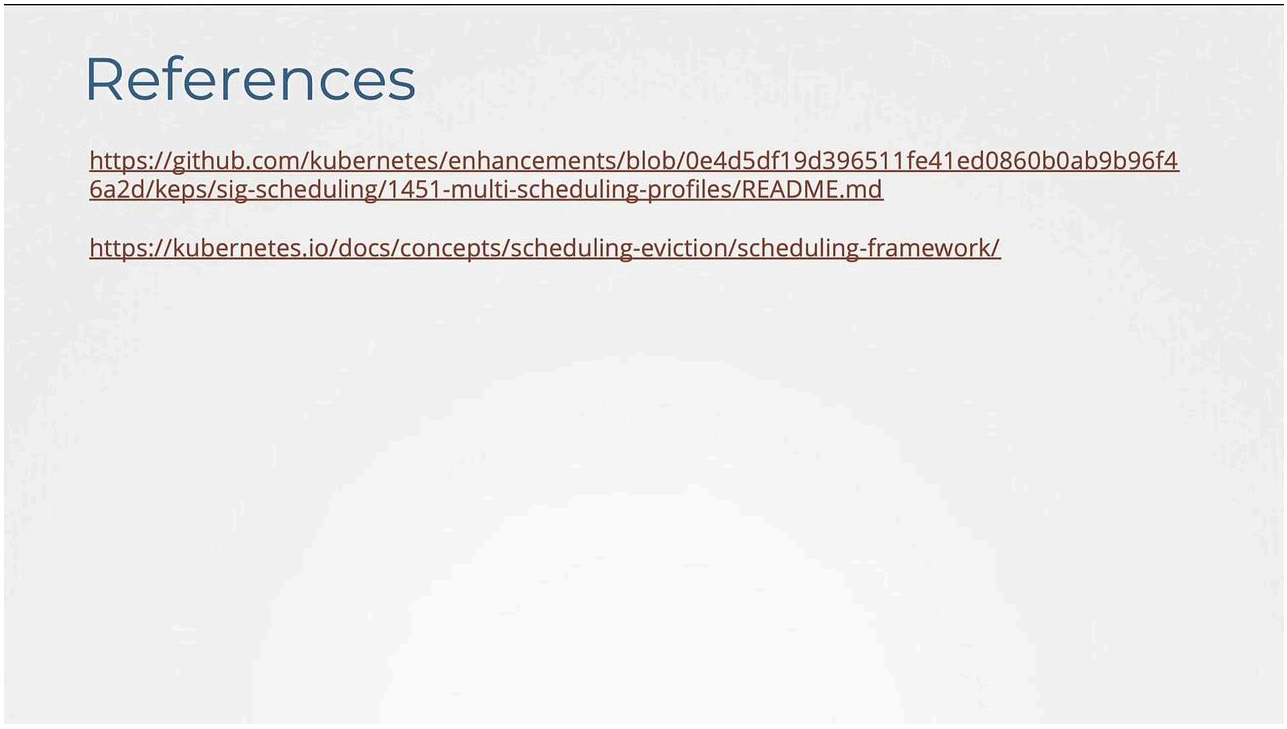
```
    score:
      disabled:
      - name: "*"

- schedulerName: my-scheduler-4
```

Under the plugins section for each profile, you can specify which extension points to modify and choose to selectively enable or disable plugins by name or using a pattern.

Additional Resources

For more information on multi-scheduling profiles, refer to the **Kubernetes enhancement proposal CAP-1451** and other related scheduling framework articles.



That concludes our overview of configuring Kubernetes scheduler profiles. Happy scheduling!

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